Object Grasping using Kinova Robot Arm

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Project motivation/ Goal

Motivation:

- Traditional robotic systems struggle with dynamic and precise object handling in cluttered environments.

- Real-time visual feedback offers a way to enhance robot effectiveness and adaptability.

Goal:

- **Develop a robotic environment system capable of:**
  - Detecting/Segment objects in real-time.
  - Dynamically adjusting its position using visual feedback.
  - Accurately grasping and relocating objects.

- **Utilize ROS2/Isaac-ROS and Gazebo/Isaac-Sim for integrating all the components.**

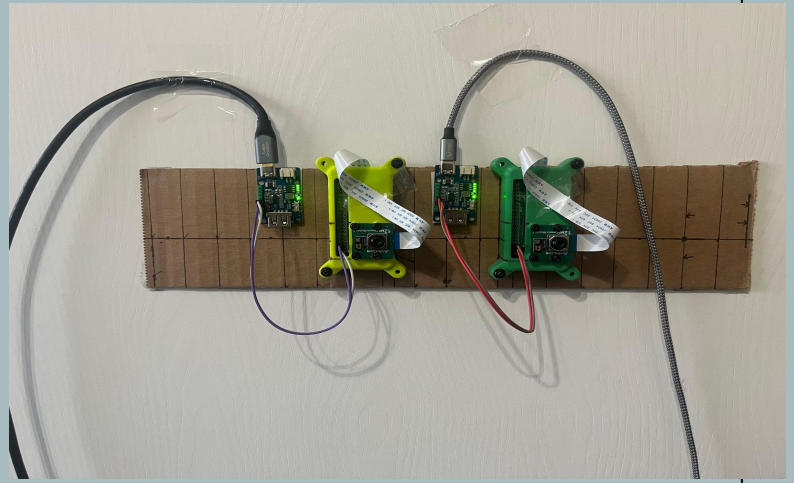
Approach/solution

Core Approach:

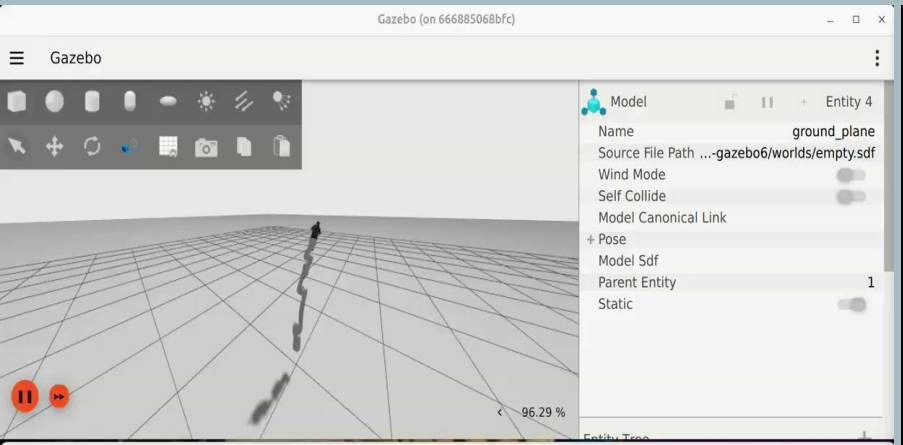
- Leveraging Image-Based Visual Servoing (IBVS) along with a robotic arm to enable precise grasping and movement of objects positioned on a table.

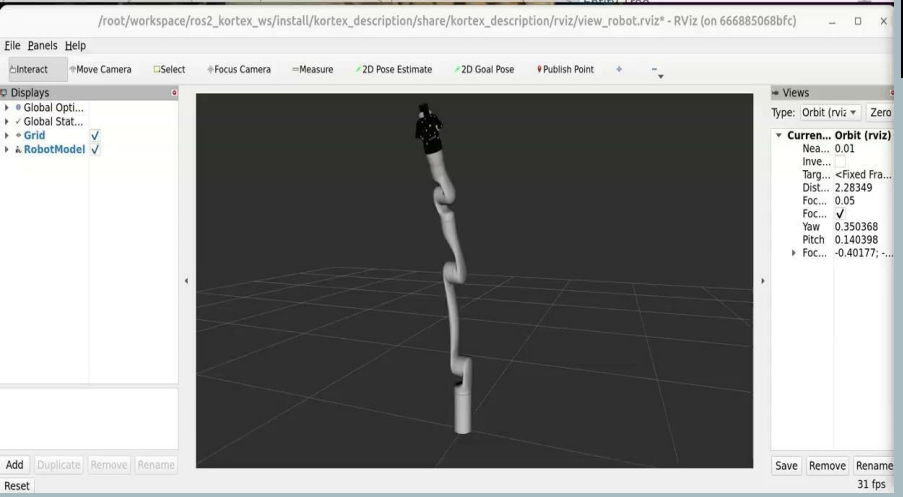
Step-by-Step Process:

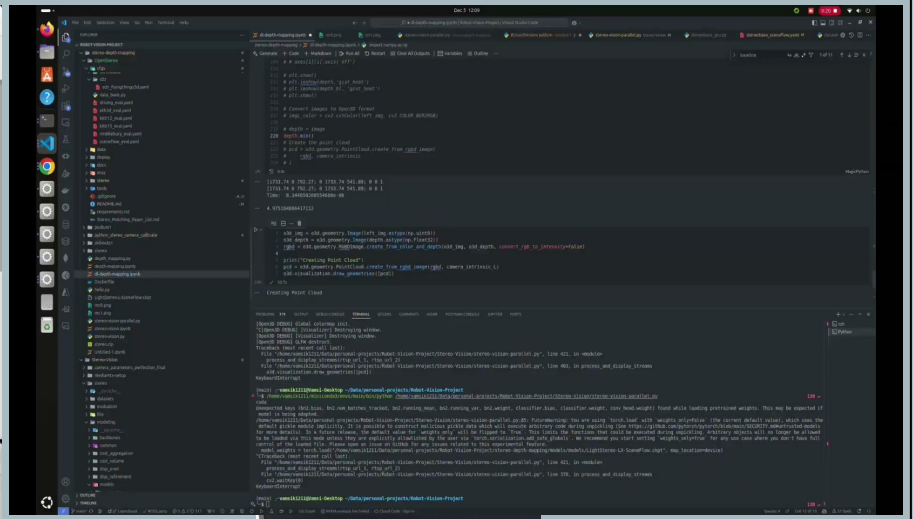
- **Visual Input:** Used a Stereo Camera for continuous depth and RGB data.
- **Segmentation and Localization:** Segment desired objects and create a real-time 3D point cloud of their positions to estimate depth.
- **Real-Time Feedback Loop:** Process visual data to guide the robotic arm dynamically i.e. loop processing should be below 30ms per frame.
- **Robotics Arm Control:** Calculate precise inverse kinematics using depth information, and execute controlled grasping. Publish depth/position data to robotic arm using ROS2 for placement and grasping.

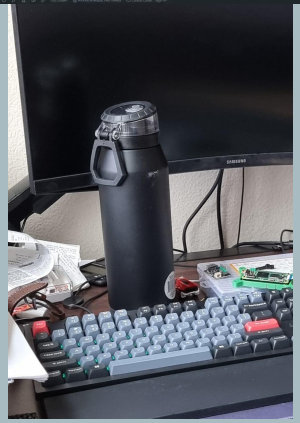


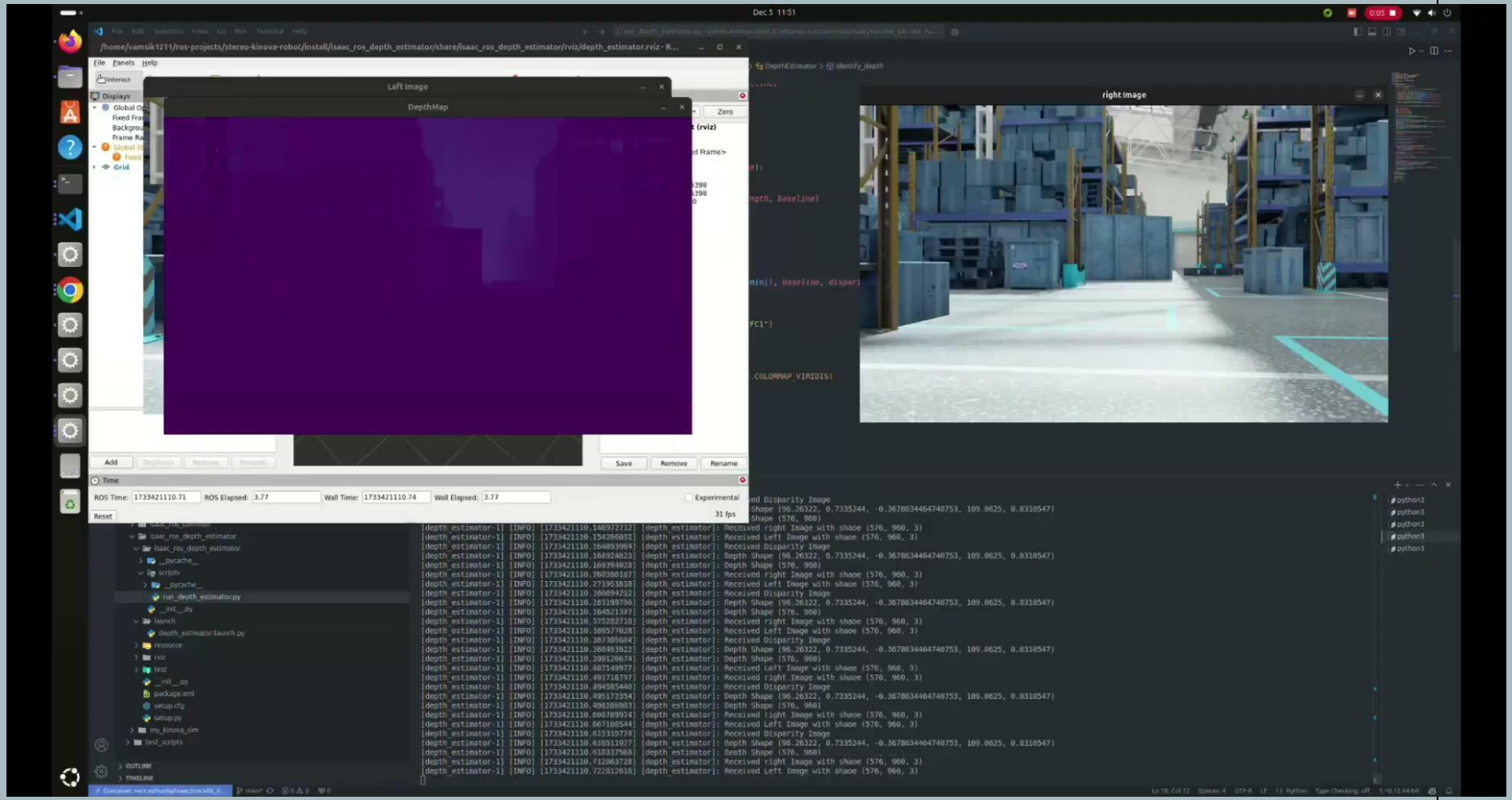
Results/demo







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Left Image
DepthMap
right Image
Time
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ROS Elapsed: 3.77
Wall Time: 1733421110.74
Wall Elapsed: 3.77
Experimental
Reset
31 fps
Save
Remove
Rename
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Q&A
Any Questions?